LECTURE 23

21 November 2023 11:34

$$f(x,y) = x^4 y^3$$

$$f_x = 4x^3 y^3 \qquad f_y = 3x^4 y^2$$

1) $f_{xx} = 4y^3 \cdot 3x^2 = 12x^2 y^3$ 2) $f_{yx} = 4x^3 \cdot 3y^2 = 12x^3 y^2$

3) $f_{xy} = 3y^2 \cdot 4x^3 = 12x^3 y^2$ 4) $f_{yy} = 3x^4 \cdot 2y = 6x^4 y$

Hessian matrix $H(x,y) = \begin{bmatrix} f_{xx} & f_{yx} \\ f_{xy} & f_{yy} \end{bmatrix}$

$$D(x,y) = |H(x,y)| = f_{xx} f_{yy} - (f_{xy})^2$$

Remarks: f_{xy} & f_{yx} are not equal in general.

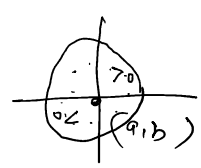
Maxima & minima in several variables:

Critical points : $f_x = 0$ & $f_y = 0$ simultaneously (x,y)

Two types: 1) Extreme pts 2) Saddle points

1) $\Rightarrow$ either $f(a,b) < f(x,y)$ or $f(a,b) > f(x,y)$ for all point (x,y) in the nbd of (a,b)

2) $\Rightarrow$ if at some points f^n al value is less and at some other points f^n al value is more in nbd of (a,b)



Test:

1) $D(x,y) > 0$ $\begin{cases} f_{xx} > 0 & \text{local minima} \\ f_{xx} < 0 & \text{local maxima} \end{cases}$ $f(a,b) = 0$

2) $D(x,y) < 0$ Saddle points.

3) $D(x,y) = 0$ Further investigation needed (no conclusion)

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1) $f(x,y) = 3x^3 - \frac{9}{2}x^2 + 15y^3 - \frac{45}{2}y^2 + 17$

$f_x = 9x^2 - 9x = 9x(x-1) = 0 \Rightarrow x=0$ or $x=1$

$f_y = 45y^2 - 45y = 45y(y-1) = 0 \Rightarrow y=0$ or $y=1$

Critical points : $(0,0), (0,1), (1,0), (1,1)$

$D(x,y) = f_{xx} f_{yy} - (f_{xy})^2 = 405\,(2x-1)(2y-1)$

$f_{xx} = 18x - 9 = 9(2x-1)$

$f_{yy} = 90y - 45 = 45(2y-1)$ $\qquad f_{xy} = 0$

points	$D(x,y)$	Inference	f_{xx}
$(0,0)$	>0		$-9<0$ Max
$(0,1)$	<0	saddle	
$(1,0)$	<0	saddle.	
$(1,1)$	>0		$9>0$ Min.

2) $f(x,y) = x^3 + y^3$

critical points : $f_x = 0$ & $f_y = 0$

$3x^2 = 0$ $\qquad 3y^2 = 0$ $\qquad (0,0)$

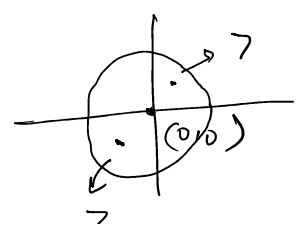
$x = 0$ $\qquad y = 0$ $\qquad$ saddle point.

$D(x,y) = f_{xx} f_{yy} - (f_{xy})^2 = 6x \cdot 6y - 0 = 36xy$

$D(0,0) = 0$ (No conclusion) Further investigation.

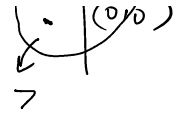
In I st quadrant : $x>0, y>0$ $\quad x^3+y^3>0$

$f(x,y) > 0$



III quad. $x<0, y<0$ $\quad x^3+y^3<0$

III quad. $x < 0, y < 0$ $x^3 + y^3 < 0$



3) $f(x,y) = xy - x^2 - y^2 - 2x - 2y + 4$